URBAN BOUNDARY LAYER

Academic English Vocabulary Test

城市边界层学术英语词汇自测卷

试卷·答案·逐题解析

15 道词汇选择｜ 1 篇完形填空｜ 2 篇阅读理解｜ 5 道语境填空

重点检测：词义辨析·固定搭配·语境推断·学术阅读

姓名		日期	
建议用时	60 分钟	满分	100 分
核心材料	*Progress in observing and modelling the urban boundary layer*		

论文语境核对。

DIGITISED VOCABULARY LIST

原图内容数字化重编码

下表按两张图片的顺序重新编码，共 4 个基础词与 29 个论文短语。中文义经过论文摘要语境核对；“用法与易错点”是为本次自测补充的规范化说明，并非对原图内容的机械扩写。

A. General Words •基础词

编号	English	推荐中文义	用法与易错点
A1	***progress***	进展；进步	通常作不可数名词：***make progress in/towards***；不写 ***a progress*** 或 ***many progresses***。
A2	***model***	v. 建模；模拟 n. 模型	***model airflow***、***a numerical model***。本文是科研义，不是“模范”。英式拼写 ***modelling***，美式拼写 ***modeling***。
A3	***boundary***	边界；界限	***the boundary between A and B***；在专业复合词中为 ***boundary layer***，不宜机械替换为 ***border***。
A4	***layer***	层	***a layer of air***、***mixed layer***、***residual layer***；注意与 ***level***（高度/水平）区分。

B. Article Expressions •论文词汇与短语

编号	English expression	推荐中文义	用法、结构与易错点
P1	***urban boundary layer***	城市边界层（UBL）	指直接响应城市地表强迫的低层大气；不是“城市边缘层”。
P2	***given***	鉴于；考虑到	本文为介词/连词用法：***given*** + 名词短语；***given that*** + 从句。也可意为“给定的”，但不是本文义。
P3	***develop cities sustainably***	以可持续方式发展城市	***sustainably*** 是副词，修饰 ***develop***。***develop sustainable cities*** 语法正确，但信息重心略有不同。
P4	***sound***	可靠的；合理的；有充分依据的	***sound evidence/methods/reasoning***。本文不是“有声的”；可用 ***reliable***、***well-founded*** 释义。
P5	***become pressing***	变得日益迫切	***pressing*** 意为 ***urgent***；常见搭配还有 ***an increasingly pressing need***。
P6	***This review paper considers ...***	本综述考察/探讨……	综述写作中的 ***consider*** 接近 ***examine***；主语 ***paper*** 为单数，动词用 ***considers***。
P7	***progress made in studies of ...***	在……研究中取得的进展	***made in studies of ...*** 是过去分词短语，后置修饰 ***progress***；可展开为 ***progress that has been made ...***。
P8	***in terms of a conceptual framework***	依据一个概念框架；从概念框架角度	固定形式是复数 ***terms***；后接名词或动名词。此处强调综述用该框架组织讨论。
P9	***spanning A to B***	涵盖/跨越从 A 到 B 的范围	***span*** 表示覆盖范围；也可自然改写为 ***ranging from A to B***，不应把该同义表达判为错误。

编号	English expression	推荐中文义	用法、结构与易错点
P10	***microscale to mesoscale***	从微尺度到中尺度	作整体前置定语时可写 ***microscale-to-mesoscale processes***。
P11	***determinants***	决定因素；控制因素	***determinants of UBL structure***。勿混淆 ***determination***（决心/确定）与 ***determiner***（限定词）。
P12	***considerable***	相当大的；显著的	***considerable*** + 名词，如 ***considerable progress/depth***；***considerably*** 是副词。
P13	***urban roughness sub-layer***	城市粗糙子层（RSL）	建筑尾流仍直接影响平均流动的城市近地层。***sub-layer*** 与 ***sublayer*** 均可见，不以拼写差异判断知识对错。
P14	***assumptions about atmospheric turbulence***	关于大气湍流的假设	***make/test/challenge assumptions about …***；也可用 ***the assumption that*** + 从句。
P15	***break down***	失效；不再成立/适用	本文是不及物用法：***the assumptions break down***。***break something down*** 另可表示“分解、详细解释”。
P16	***dominate***	主导；支配	***A dominates B***；被动式 ***B is dominated by A***。通常不需要冗余的 ***dominate over***。
P17	***coupling of A to B***	A 与 B 之间的耦合/联系	原文是地表与 UBL 的耦合；更一般的 ***coupling between A and B*** 也很自然。
P18	***residual layer***	残留层；余留层	日落后保留在浅薄近地稳定层上方的原混合空气；不是“残差层”。勿混 ***resident/resilient***。
P19	***remain under-researched***	仍研究不足；仍缺乏充分研究	***remain*** + 形容词/过去分词；***remain understudied*** 是可接受的近义表达。
P20	***remote sensing methods***	遥感方法	通过非接触信号获取目标信息，如 lidar。***remote-sensing methods*** 也可作为编辑风格。
P21	***high-resolution modelling tools***	高分辨率建模工具	***high-resolution*** 作复合定语通常加连字符；***modelling/modeling*** 分别为英式/美式拼写。
P22	***permit rapid progress***	使快速进展成为可能；促进快速进展	***permit*** + 名词；也可写 ***permit somebody to make rapid progress***。***progress*** 不可数。
P23	***surface heterogeneity***	地表/下垫面异质性	指材料、植被、形态、热释放等空间差异。形容词为 ***heterogeneous***。
P24	***be strongly considered***	得到更多/更充分重视	原文完整语境为 ***should be more strongly considered***。自由写作中 ***receive greater attention*** 或 ***be more fully considered*** 往往更自然。
P25	***specific research priorities***	具体研究重点/优先事项	***identify/set research priorities***；***specific*** 是“具体的”，不等于 ***special***。
P26	***urban canopy***	城市冠层；建筑与街谷构成的近地城市空间	不只是植物冠层。***urban canopy layer*** 常指约平均建筑高度以下的层。
P27	***high-rise***	高层的；高层建筑	***high-rise buildings***、***high-rise urban canopies***；作定语通常保留连字符。
P28	***spatially extensive***	空间覆盖广泛的	***spatially extensive measurement networks***；副词 ***spatially*** 修饰形容词 ***extensive***。
P29	***be coupled to***	与……相耦合；与……紧密联系	***be strongly coupled to model development***。***coupled with*** 在其他句法语境中也可能正确。

C. Coverage Map •词条覆盖图

每个核心词条都至少在题干、完形、阅读或主动填空中出现；关键搭配在解析中再次复现。下表给出最主要的检测位置，便于按错题反查。

词条组	主要题号	主要能力
progress; considerable; permit rapid progress	1, 15, 37–38	可数性、词性、结构迁移
model; boundary; layer; urban boundary layer	11–12, 18, 23, 31	专业义、定义识别
given; develop cities sustainably; become pressing	2, 4, 36	句首结构、副词位置、语境义
sound; This review paper considers	3, 30, 33, 37	一词多义、综述写作
in terms of; conceptual framework; spanning A to B; microscale to mesoscale	5–6, 16–17, 39	固定介词结构、范围表达
determinants; dominate; surface heterogeneity	7, 9, 13, 21–22	因果词义、词形辨析
urban roughness sub-layer; assumptions about atmospheric turbulence; break down	8, 20, Passage A	专业搭配、短语动词
coupling of A to B; be coupled to	10, 40	名词结构、被动结构
residual layer; remain underresearched	23, 31–32	定义、细节定位
remote sensing methods; high-resolution modelling tools	24–25, 34	复合名词、方法辨析
be strongly considered; specific research priorities	28, Passage B	语境改写、学术表达
urban canopy; high-rise; spatially extensive	14, 18–19, Passages A–B	复合词、词性、范围含义

REVIEW NOTES

复习区核验说明

编制边界 两张图片中的词条保持原顺序重新编码；释义按论文语境校正。试卷中的完形与两篇阅读均为原创，不复制论文摘要或既有考试题。专业背景只用于提供自然语境，答案均可从英语表达或文内信息推出。

核验层级

1. **原始词条：**用户提供的 Memo No. 131 与 Memo No. 132 图片。
2. **论文语境：**Janet F. Barlow, *Progress in observing and modelling the urban boundary layer*, *Urban Climate*, 10(2), 216–240 (2014), DOI 10.1016/j.uclim.2014.03.011； University of Reading repository。
3. **词义与搭配：**Cambridge Dictionary的 sound、progress、pressing 词条用于核对学术义、不可数性和紧迫义。
4. **技术术语边界：**American Meteorological Society: roughness sublayer 与 NOAA: Doppler lidar 用于核对粗糙子层与遥感例子的基本表述。

两个值得特别记住的规范化点

- 原论文采用英式拼写 ***modelling***；美式 ***modeling*** 同样正确，不应把拼写体系差异误判成词汇错误。
- ***should be more strongly considered*** 在原语境中合法，意为“应得到更多重视”；自由写作中可用更自然的 ***should receive greater attention***。

复习结束·请从下一页开始独立作答

完成全部题目后，再翻到整份 PDF 的最后一部分核对答案。

作答说明

- 全卷共 40 题，满分 100 分。除第四部分外，每题只有一个最佳答案。
- 先独立完成试卷，再查看解析版。选择题请在题号旁写出 A、B、C 或 D。
- 阅读材料为原创文本；不要求具备城市气象学专业背景，所有答案均可由语言或文意推出。

PART Ⅰ Vocabulary & Collocation

30 分

Directions: Choose the option that best completes the sentence or explains the underlined expression. *(15 questions, 2 points each)*

1. 基础 Researchers have made _____ progress in observing the urban atmosphere.

A. considerate
B. considerable
C. considering
D. considered

2. 基础 _____ the limited number of long-term observations, the team avoided making a universal claim.

A. Giving
B. Given
C. Give
D. To giving

3. 基础 In the phrase "***sound*** modelling and observational tools," ***sound*** most nearly means:

A. producing a loud noise
B. financially profitable
C. reliable and well-founded
D. simple enough for beginners

4. 基础 If a need ***becomes pressing***, it:

A. becomes increasingly urgent
B. is gradually forgotten
C. becomes physically heavier
D. has already been satisfied

5. 中等 Which sentence uses ***in terms of*** most naturally?

A. In terms of the instrument broke, the team stopped.
B. The two methods differ in terms of spatial coverage.
C. The report is in terms of five pages.
D. The model in terms of predicts tomorrow.

6. 中等 A framework ***spanning the microscale to the mesoscale*** covers:

A. only processes smaller than a building
B. only regional weather systems
C. a range extending from very local to city-regional scales
D. two unrelated periods of historical time

7. 基础 In "the main determinants of boundary-layer structure," ***determinants*** are:

A. measuring devices
B. controlling factors
C. final conclusions
D. geometric boundaries

8. 中等 If assumptions about atmospheric turbulence ***break down*** near tall buildings, the assumptions:

A. become more detailed
B. are divided into smaller assumptions
C. cease to be valid in that setting
D. are translated into another language

9. 基础 Surface heterogeneity may ***dominate*** the observed pattern. This means it may:

A. have the strongest influence on the pattern
B. disappear from the pattern
C. measure the pattern remotely
D. make the pattern perfectly uniform

10. 中等 The ***coupling of the surface to the urban boundary layer*** refers to:

A. the physical interaction linking the surface and the air above it
B. the distance between two independent instruments
C. the boundary separating a city from the countryside
D. the pairing of two words with the same meaning

11. 基础 In this study, to ***model*** airflow means to:

A. present a fashionable example for other cities to copy
B. represent or simulate the physical process
C. draw only the administrative boundary of a city
D. build a small physical object and avoid all calculations

12. 进阶 The part of the lower atmosphere that responds directly to the city's surface is called the:

A. urban canopy
B. residual layer
C. urban boundary layer
D. high-rise boundary

13. 中等 ***Surface heterogeneity*** is best understood as:

A. spatial differences in materials, vegetation, shape, or heat release
B. a perfectly even surface extending across an entire city
C. an error caused only by a damaged surface sensor
D. the vertical distance from the road to a roof

14. 基础 A ***spatially extensive*** measurement network:

A. operates for one minute
B. covers a large geographic area
C. uses only expensive sensors
D. measures one variable in one room

15. 中等 Which sentence is grammatically correct and closest to the target meaning?

A. New tools permit researchers make rapid progress.
B. New tools permit to make researchers rapid progress.
C. New tools permit researchers to make rapid progress.
D. New tools permit researchers making rapidly progresses.

PART II Cloze

20 分

Directions: Read the passage and choose the best answer for each blank. *(10 questions, 2 points each)*

Building a Study Across Scales

Cities are difficult places in which to study the atmosphere. A research team starting a new project therefore needs a clear conceptual **(16)** ____. The plan should not stop at a single street: it may need to **(17)** ____ the microscale to the mesoscale.

At street level, the **(18)** ____ formed by buildings and roads creates complex flows. In districts filled with **(19)** ____ buildings, wakes from individual structures can reach far above roof level. As a result, simple assumptions about turbulence may **(20)** ____ in the urban roughness sub-layer.

The challenge changes again at larger scales. Differences among parks, concrete blocks, water, and industrial land create surface **(21)** ____. Under some weather conditions, these contrasts can **(22)** ____ the pattern recorded by a small number of stations. Night-time work is especially valuable because the **(23)** ____ above the shallow near-surface layer still remains under-researched.

To connect all these scales, the team combines **(24)** ____, including lidar, with **(25)** ____ that represent streets and neighbourhoods in greater detail. The observations are collected by a spatially extensive network and are coupled to model development. Together, these choices make the conclusions more defensible.

16. A. framework
B. determinant
C. boundary
D. priority

17. A. press
B. span
C. modelled
D. couple

18. A. residual layer
B. urban canopy
C. remote method
D. research priority

19. A. high-rise
B. long-term
C. spatially
D. conceptual

20. A. make up
B. break down
C. carry out
D. turn over

21. A. heterogeneity
B. boundary
C. progress
D. pressure

22.

A. develop
B. permit
C. dominate
D. remain

23. A. urban canopy
B. residual layer
C. microscale
D. model boundary

24. A. remote sensing methods
B. direct touching methods
C. conceptual priorities
D. high-rise assumptions

25. A. highly resolved model tools
B. high-resolution modelling tools
C. modelling high-resolution tools
D. high resolutions modelling tool

PART III Reading Comprehension

30 分

Directions: Read the two passages and choose the best answer. *(10 questions, 3 points each)*

Passage A —When One Neighbourhood Is Not Enough

Maya began her project with a weather tower in a compact city centre. Her first model treated the city as a uniform surface, and its equations worked reasonably well above open ground. Close to the buildings, however, the model repeatedly underestimated sudden changes in wind. The problem was not simply bad data. The tower stood inside the urban roughness sub-layer, where wakes from roofs and walls disrupted the flow. Assumptions developed for a flatter surface could break down there.

At first, Maya tried to improve the model by tuning one parameter. This helped on winds from the west but made the results worse on winds from the south. She then mapped the tower's surroundings and found parks, glass towers, brick blocks, and a river within a few kilometres. Their different shapes and thermal properties created strong surface heterogeneity. No single adjustment could represent every wind direction. The finding showed that surface differences should be more strongly considered rather than averaged away.

Maya redesigned the project in terms of a scale-aware conceptual framework. Street-level sensors examined individual wakes, lidar followed the atmosphere above the roofs, and several stations sampled different neighbourhoods. High-resolution modelling tools were used to connect these observations. The new design spanned the microscale to the mesoscale instead of treating one tower as a miniature version of the whole city.

The redesign did not make the city simple. It made the limits of each measurement explicit. That change was scientifically valuable: a sound study does not require every instrument to represent everything. It requires the researcher to know what each instrument can and cannot represent.

26. What is the main purpose of the passage?

A. To argue that all urban weather towers should be removed

B. To show why an urban study had to account for different scales and surfaces

C. To prove that lidar always gives better data than a weather tower

D. To explain how one parameter can represent every neighbourhood

27. Why did the first model perform poorly close to the buildings?

A. Its assumptions were not fully valid in the roughness sub-layer.

B. The tower measured temperature instead of wind.

C. The city contained no open ground at all.

D. The model was designed only for night-time conditions.

28. In the second paragraph, “should be more strongly considered” means that surface differences should:

A. receive greater attention in the study design

B. be measured only with physically stronger instruments

C. be removed before any model is tested

D. be assumed to have the same effect in every wind direction

29. The phrase “a miniature version of the whole city” is used to reject the idea that:

A. one site automatically represents all urban environments

B. city-scale models may include local observations

C. instruments have limits that should be reported

D. street-level sensors can examine individual wakes

30. Which statement would the author most likely agree with?

A. A sound study makes the scope and limits of its evidence clear.

B. More instruments automatically remove every source of uncertainty.

C. A uniform-surface model is accurate in every high-rise district.

D. Measurements are useful only when they cover the entire city.

Passage B —Studying the City After Sunset

During the day, heat from the surface helps mix air through a deep part of the urban boundary layer. After sunset, that continuous heating weakens. A shallow, more stable layer may then form near the ground, while some of the formerly mixed air remains above it. This upper air is called the residual layer. It can retain traces of the day’s heat, moisture, and pollution, so it may influence conditions the following morning.

Despite that possible influence, the residual layer remains under-researched. Many towers do not extend high enough to sample it, and balloon observations are too infrequent to describe its changes throughout every night. Remote sensing methods can fill part of the gap by providing repeated vertical profiles. Yet a single signal can be misleading. For example, the top of an aerosol-rich layer is not always the same as the top of active turbulence. A sound interpretation therefore compares several kinds of evidence rather than treating one detected boundary as certain.

Researchers are now developing spatially extensive networks in which lidars, surface stations, and occasional balloons operate together. The observations can be coupled to high-resolution models, allowing

teams to test whether the same explanation works on clear, cloudy, calm, and windy nights. This approach also helps identify specific research priorities instead of collecting data without a clear question. Understanding the night-time atmosphere has become pressing as cities seek to prepare for heat and poor air quality. Better evidence will not by itself develop cities sustainably, but it can give planners a more reliable basis for choosing and evaluating actions.

31. According to the passage, what is the residual layer?

A. A shallow stable layer that always touches the ground
B. Formerly mixed daytime air remaining above a new near-surface layer
C. The solid material left on rooftops after pollution events
D. A permanent boundary separating a city from the countryside

32. Why has the residual layer remained under-researched?

A. It exists only during unusual storms.
B. Its temperature cannot be represented by any model.
C. Common observations often lack sufficient height or time coverage.
D. Researchers have already agreed that it has no practical influence.

33. In the second paragraph, ***sound interpretation*** most nearly means:

A. an audible explanation
B. a reliable interpretation
C. a brief description
D. a popular opinion

34. Which research design does the author support?

A. Using one aerosol boundary as certain evidence of turbulence
B. Collecting more data before deciding on any research question
C. Combining several observation types with high-resolution models
D. Replacing all observations with a single city-wide model

35. What can reasonably be inferred from the final paragraph?

A. Scientific evidence can support planning, but it does not make policy choices by itself.
B. Sustainable urban development depends only on night-time temperature.
C. Once a city installs lidar, poor air quality will disappear.
D. Urban planners should wait until every uncertainty has been eliminated.

PART IV Contextual Completion

20 分

Directions: Complete each sentence with the exact English expression suggested by the Chinese cue. You may change capitalization, but do not change the core wording. *(5 questions, 4 points each)*

36. Given rapid urban growth, learning how to 以可持续方式发展城市 has become pressing.

_____

37. 本综述考察 how street design changes wind and temperature.

_____

38. The final chapter summarizes the 在夜间城市热研究中取得的进展.

_____

39. The evidence is organized 依据一个跨越微尺度到中尺度的概念框架.

_____

40. For a trustworthy result, a high-resolution model 必须与观测相耦合 rather than be used alone.

_____

交卷前自检 检查 *sustainably* 是否正确修饰动词；检查 *progress* 是否作不可数名词；检查 *in terms of* 与 *be coupled to* 的完整结构。

ANSWER KEY & EXPLANATIONS

答案与逐题解析

建议用法 先只核对答案；答错后再阅读定位线索、搭配规则与干扰项分析。若同一词条连续两次答错，请在附录词表中用自己的例句重新造句。

答案速查

部分	答案
Part I	1 B 2 B 3 C 4 A 5 B 6 C 7 B 8 C 9 A 10 A 11 B 12 C 13 A 14 B 15 C
Part II	16 A 17 B 18 B 19 A 20 B 21 A 22 C 23 B 24 A 25 B
Part III	26 B 27 A 28 A 29 A 30 A 31 B 32 C 33 B 34 C 35 A
Part IV	见第 36–40 题参考答案；大小写之外，核心短语应完整。

Part I —Vocabulary & Collocation

1 B •considerable

Considerable 是形容词，意为“相当大的、显著的”，可直接修饰不可数名词 ***progress***。固定搭配是 ***make considerable progress in doing something***。

A ***considerate*** 表示“体贴的”；C ***considering*** 是介词、连词或现在分词；D ***considered*** 常表示“经过深思熟虑的”。三者都不能在此表达“显著的进展”。

例：The team made considerable progress in modelling street-level wind.

2 B •Given

句首 ***Given*** + **名词短语**表示“鉴于、考虑到”，整体作原因或条件状语。本句可改写为 ***Considering the limited number of observations***，但在给定选项中只有 B 的形式正确。

A ***Giving*** 表示“给予”；C 是动词原形，不能这样独立引出状语；D 中 ***to giving*** 结构错误。

例：Given the uncertainty, the conclusion should remain cautious.

3 C •reliable and well-founded

Sound 在学术语境中常表示“可靠的、合理的、有充分依据的”，如 ***sound evidence***、***sound methods*** 和 ***sound reasoning***。这里与“声音”无关，也不表示工具一定盈利或只适合初学者。

例：Policy decisions should be based on sound observational evidence.

4 A •becomes increasingly urgent

Pressing 表示“紧迫的、亟待处理的”。***A need becomes pressing*** 即“某种需要变得越来越迫切”。它不是“被按压”，也不表示需求已经得到满足。

例：During a heat wave, the need for reliable forecasts becomes pressing.

5 B

In terms of **+ 名词/动名词**表示“就……而言、从……角度看”。B 中 ***spatial coverage*** 是完整名词短语，句义为“两种方法在空间覆盖方面不同”。
A 应改为 ***Because the instrument broke***；C 中页数应直接说 ***The report is five pages long***；D 把短语插进主谓之间，句法不成立。

6 C

Span 表示“横跨、涵盖一个范围”。***spanning the microscale to the mesoscale*** 因而指从非常局地的建筑/街道尺度，一直覆盖到城市或区域尺度。A、B 都把范围错误地缩成单一端点；D 把空间尺度误解成历史时期。
同义改写：covering scales from the microscale to the mesoscale.

7 B •controlling factors

Determinants of X 是“决定或控制 X 的因素”。它与 ***determination***（决心；确定）及 ***determiner***（语法中的限定词）不同。题干中的 ***boundary*** 也不是 ***determinant*** 的同义词。
例：Surface heating and wind are important determinants of layer depth.

8 C •cease to be valid

Break down 在本语境中是不及物短语动词，表示理论、假设或系统“失效、不再成立”。A 与原意相反；B 是 ***break something down*** 的另一种用法，表示“分解/逐项解释”；D 与语言转换无关。
例：The uniform-flow assumption may break down near tall buildings.

9 A

Dominate 是及物动词，表示“支配、主导、产生最强影响”，可写 ***A dominates B*** 或 ***B is dominated by A***。B、D 与“主导”相反；C 把影响因素误写成了测量手段。
例：Local traffic emissions dominated the morning concentration pattern.

10 A

Coupling of A to B 强调 A 与 B 之间存在可传递影响的物理联系。在城市边界层语境中，地表的热量、动量与水汽交换会影响上方空气，空气状态也会反过来影响地表交换。其余选项分别描述距离、行政边界和词义关系，不是物理耦合。

11 B •represent or simulate

Model 作动词时表示“为……建立模型、模拟”，即用物理、数学或计算表示来研究真实过程。B 准确表达了这一专业义。
A 把 ***model*** 误解为“作榜样”；C 只涉及行政边界，并没有模拟气流；D 把模型限制成实体缩尺物，而且“避免所有计算”与题干研究语境冲突。
例：The software models airflow around a group of high-rise buildings.

12 C •urban boundary layer

Urban boundary layer（UBL）是受到城市地表摩擦、加热、储热、蒸散等作用直接影响的低层大气。题干的定义线索是 ***lower atmosphere***、***responds directly*** 和 ***city's surface***。
A ***urban canopy*** 主要指建筑、街谷等构成的近地城市空间；B ***residual layer*** 是日落后保留在浅薄近地层上方的原混合空气；D 不是标准术语。

13 A

Heterogeneity 是“异质性、不均一性”。城市地表可在材料、植被、建筑形态、水体和热排放上存在空间差异，所以 A 是完整解释。B 定义的是均一性；C 将真实空间差异误说成传感器故障；D 只是一个高度差。
词形：heterogeneity（名词）—heterogeneous（形容词）。

14 B

Spatially 是副词“在空间上”，修饰 ***extensive***“范围广的”。因此该短语强调网络覆盖较大的地理区域，不表示持续时间、价格或室内单点测量。
例：A spatially extensive network can compare different neighbourhoods.

15 C

Permit 有两个与本题相关的正确框架：***permit*** + **名词**（***permit rapid progress***）和 ***permit somebody to do something***。C 完整使用第二种结构。
A 缺少 ***to***；B 词序错误；D 应用 ***progress*** 的不可数形式，且 ***permit somebody doing*** 不是本句要表达的结构。

Part II —Cloze

16 A •framework

Conceptual framework 是固定学术搭配，意为“概念框架”。它为研究者组织尺度、变量与关系提供结构。其他三个词虽都能单独出现，但不能自然构成 ***a clear conceptual ...*** 并承接下一句的研究范围。

17 B •span

Need to 后接动词原形；***span A to B*** 表示“涵盖从 A 到 B”。A ***press*** 与尺度范围无关；C 是过去分词；D ***couple*** 通常需要说明被连接的两个对象，而不是表示覆盖范围。

18 B •urban canopy

后面的 ***formed by buildings and roads*** 是定位线索：建筑、街谷和道路共同构成 ***urban canopy***。残留层位于夜间近地层之上；另外两项不表示这种城市近地空间。

19 A •high-rise

High-rise buildings 是“高层建筑”的标准复合表达。B 描述持续时间；C 是副词，不能单独修饰 ***buildings***；D ***conceptual*** 与建筑高度无关。作前置定语时 ***high-rise*** 通常保留连字符。

20 B •break down

句意是建筑尾流使简单湍流假设“不再适用”。***May*** 后必须接动词原形；***break down*** 在此语义与句法都匹配。A 是“组成”；C 是“实施”；D 是“翻转/周转”，均不合逻辑。

21 A •heterogeneity

公园、混凝土、水面和工业用地之间的差异构成 ***surface heterogeneity***。句中需要名词作 ***surface*** 后的中心词。B 只表示界线，C 是进展，D 是压力，都不能概括这些空间差异。

22 C •dominate

主语 ***these contrasts*** 可“主导”少数站点记录到的格局。***Can*** + **动词原形**的结构满足所有动词选项，但只有 ***dominate*** 表达“产生最强影响”。A“发展”、B“允许”、D“保持”在语义上均不完整。

23 B •residual layer

Above the shallow near-surface layer 是关键定位：残留层位于新形成的浅薄近地层上方，并且原文语境明确指出其 ***remains under-researched***。城市冠层位于建筑附近；微尺度是一类尺度；D 不是标准术语。

24 A •remote sensing methods

Including lidar 直接给出例子。激光雷达是遥感方法，因此 A 唯一匹配。B 不是标准研究分类；C、D 都把不相干的词机械拼接，语义不成立。

25 B •high-resolution modelling tools

复合定语 ***high-resolution*** 通常加连字符，***modelling*** 修饰 ***tools***，词序为“高分辨率 + 建模 + 工具”。A 可在别的语境中描述“被高度解析的工具”，但不是目标短语；C 词序不自然；D 的单复数与复合修饰均错误。

Part III —Reading Comprehension

26 B

定位：第一段指出均一表面假设在建筑附近失效；第二段说明不同地表造成异质性；第三段给出跨尺度的新设计。三段共同服务于 B。
A 过度推断，文章没有主张拆除塔；C 把“组合使用”夸大成 lidar 永远更好；D 与调参对不同风向产生相反效果的事实矛盾。

27 A

定位：第一段末句明确说，为较平坦地表建立的假设在城市粗糙子层中可能 ***break down***。这不是单纯的数据损坏，正是 A。
B、D 在文中没有依据；C 的绝对词 ***no open ground at all*** 还与 ***above open ground*** 的描述冲突。

28 A

Be more strongly considered 在这里不是字面上的“更用力地思考”，而是“在研究设计、数据解释和模型表示中得到更多/更充分的重视”，所以 A 正确。
B 把 ***strongly*** 错解为仪器的物理强度；C 与研究异质性的目的相反；D 正是文章批评的均一化假设。在自由学术写作中，更自然的同义表达包括 ***receive greater attention*** 和 ***be more fully considered***。

29 A

Treating one tower as a miniature version of the whole city 是形象化表达，指错误地假设单站能自动代表整个城市。新设计加入不同社区和不同尺度，正是在修正这一假设。B、C、D 都是文章支持而非否定的做法。

30 A

末段直接给出判断标准：可靠研究应知道每件仪器“能代表什么、不能代表什么”。因此 A 准确概括作者立场。B 含 ***automatically/every***，超出证据；C 与第一段冲突；D 错误地把局地但有明确代表范围的测量判为无用。

31 B

定位：第一段说明，日落后近地面形成新层，而一部分原来混合的空气保留在其上方，这就是残留层。A 把上下位置颠倒；C 把大气层误作屋顶固体残留；D 把随天气演变的大气结构误作永久行政边界。

32 C

定位：第二段给出两个互补原因：塔往往不够高，气球观测又不够频繁。前者是高度覆盖不足，后者是时间覆盖不足，所以 C 最完整。A、B、D 都使用文章未支持的绝对说法。

33 B •a reliable interpretation

此处 ***sound*** 仍是学术义“可靠的、有依据的”。下一句提出比较多种证据，说明可靠性来自交叉核对。A 误用普通名词“声音”的联想；C“简短”和 D“受欢迎”都不能解释为何要比较多种证据。

34 C

第三段明确支持将 lidar、地面站和气球组成空间覆盖广的网络，并把观测与高分辨率模型相耦合。A 与“单一信号可能误导”冲突；B 与“识别具体研究重点”冲突；D 则错误地用模型取代全部观测。

35 A

末段使用了审慎结构：更好的证据“不能单独”实现可持续城市发展，但能为规划者选择和评价行动提供更可靠的基础。因此 A 保留了科学证据与政策决策之间的边界。B、C 把多因素问题缩成单一变量或仪器；D 要求消除全部不确定性，文中没有支持。

Part IV —Contextual Completion

36 develop cities sustainably

完整句：Given rapid urban growth, learning how to develop cities sustainably has become pressing.
How to **+ 动词原形后用** ***develop***；副词 ***sustainably*** 修饰动作，回答“以怎样的方式发展”。***Develop sustainable cities*** 在语法上也成立，但重心变成“发展具有可持续属性的城市”，不完全等同于本题指定表达。
迁移句：Reliable evidence can help communities develop cities sustainably.

37 This review paper considers

完整句：This review paper considers how street design changes wind and temperature.
Consider 在综述写作中可表示“考察、探讨、审视”，接名词、名词性从句或疑问词引导的从句。本题主语 ***paper*** 为第三人称单数，所以用 ***considers***，不是 ***consider***。
迁移句：This review paper considers progress made in urban observations.

38 progress made in studies of

完整句：The final chapter summarizes the progress made in studies of night-time urban heat.
Progress 是中心名词；***made in studies of ...*** 是过去分词短语，后置修饰 ***progress***，相当于 ***progress that has been made in studies of ...***。这里不能写 ***a progress*** 或 ***progresses***。
结构对照：Researchers made progress in studying X（主动）；progress made in studies of X（名词 + 后置修饰）。

39 in terms of a conceptual framework spanning the microscale to the mesoscale

完整句：The evidence is organized in terms of a conceptual framework spanning the microscale to the mesoscale.
In terms of **+ 名词短语**表示“依据……、从……角度”。后面的 ***spanning ...*** 是现在分词短语，修饰 ***framework***；***the microscale to the mesoscale*** 给出覆盖范围。固定结构是复数 ***terms***，不能写 ***in term of***。

40 must be coupled to

完整句：For a trustworthy result, a high-resolution model must be coupled to observations rather than be used alone.
Must 后接动词原形 ***be***；***coupled*** 是过去分词，形成被动结构 ***be coupled to***，意为“与……相耦合/紧密联系”。本句强调模型与观测之间需要相互检验，而不是用模型替代证据。
用法边界：***Coupled with*** 在其他语境也可能自然，但本题要求复现图片中的 ***be coupled to***。

分数解释（建议） 90–100：词义、搭配与阅读迁移均较稳；75–89：整体掌握，建议复习错题对应短语；60–74：能识别大意，但固定结构仍需主动造句；低于 60：先熟记附录词表，再隔 24–48 小时重做纯试卷版。

错因复盘与二次测试

错因类型	典型信号	立即采取的复习动作
词义混淆	把 ***sound***、***model***、***pressing*** 按普通义理解	写出“本文义 + 一个不属于本文的普通义”，再各造一句。
词性/词形	混淆 ***considerable/considerably***、***spatial/spatially***	圈出被修饰成分，判断空格需要形容词还是副词，再重写完整短语。
固定搭配	漏掉 ***in terms of*** 的 ***-s***，或漏掉 ***permit somebody to do*** 中的 ***to***	把正确结构遮住，按“框架 + 自己的新例句”默写两遍。
可数性	写出 ***a progress*** 或 ***rapid progresses***	将 ***progress*** 作为不可数名词，与 ***make/much/considerable/rapid*** 组成四个搭配。
阅读推断	选择含 ***always/every/only/prove*** 的过强选项	在原文旁写“原文说到哪里为止”，再检查选项是否扩大了范围或因果。
专业概念	混淆 UBL、urban canopy、roughness sub-layer 与 residual layer	按“地表/建筑附近—整个受城市影响的低层大气—夜间上方残留空气”重画层次关系。

二次测试建议

1. 当天只订正错题，并为每个错词写一个与原题不同的新句子；
2. 24–48 小时后使用“纯试卷版”重新作答，不先看词汇复习表；
3. 将连续两次答错的词条加入个人清单，一周后只重测这些词条。